

1. The net dynamic force acting on the shaft is equal to zero. This requires that the line of action of three centrifugal forces must be the same. In other words, the centre of the masses of the system must lie on the axis of rotation. This is the condition for *static balancing*.

2. The net couple due to the dynamic forces acting on the shaft is equal to zero. In other words, the algebraic sum of the moments about any point in the plane must be zero.

The conditions (1) and (2) together give *dynamic balancing*.

16.11 SENSITIVENESS OF A GOVERNOR

A governor is said to be sensitive when it readily responds to a small change of speed. The movement of the sleeve for a fractional change of speed is the measure of sensitivity.

As a governor is used to limit the change of speed of the engine between minimum to full-load conditions, the sensitiveness of a governor is also defined as the ratio of the difference between the maximum and the minimum speeds (range of speed) to the mean equilibrium speed. Thus,

$$\begin{aligned}\text{Sensitiveness} &= \frac{\text{range of speed}}{\text{mean speed}} \\ &= \frac{N_2 - N_1}{N} = \frac{2(N_2 - N_1)}{N_1 + N_2}\end{aligned}\quad (16.10)$$

When N = mean speed

N_1 = minimum speed corresponding to full-load conditions

N_2 = maximum speed corresponding to no-load conditions

16.12 HUNTING

Sensitiveness of a governor is a desirable quality. However, if a governor is too sensitive, it may fluctuate continuously, because when the load on the engine falls, the sleeve rises rapidly to a maximum position. This shuts off the fuel supply to the extent to affect a sudden fall in the speed. As the speed falls to below the mean value, the sleeve again moves rapidly and falls to a minimum position to increase the fuel supply. The speed subsequently rises and becomes more than the average with the result that the sleeve again rises to reduce the fuel supply. This process continues and is known as *hunting*.

16.13 ISOCHRONISM

A governor with a range of speed zero is known as an *isochronous governor*. This means that for all positions of the sleeve or the balls, the governor has the same equilibrium speed. Any change of speed results in moving the balls and the sleeve to their extreme positions. However, an isochronous governor is not practical due to friction at the sleeve.

For a Porter governor, with all arms equal in length and intersecting on the axis (neglecting friction),

$$h_1 = \frac{g}{\omega_1^2} \left(1 + \frac{M}{m} \right) \quad \text{and} \quad h_2 = \frac{g}{\omega_2^2} \left(1 + \frac{M}{m} \right)$$

For isochronism, $\omega_1 = \omega_2$ and thus $h_1 = h_2$. However, from the configuration of a Porter governor, it can be judged that it is impossible to have two positions of the balls at the same speed. Thus, a pendulum type of governor cannot possibly be isochronous.

In the case of a Hartnell governor (neglecting friction),

At ω_1 ,

$$mr_1\omega_1^2a = \frac{1}{2}(Mg + F_{s1})b$$

At ω_2 ,

$$mr_2\omega_2^2a = \frac{1}{2}(Mg + F_{s2})b$$

For isochronism, $\omega_1 = \omega_2$.

$\therefore$

$$\frac{Mg + F_{s1}}{Mg + F_{s2}} = \frac{r_1}{r_2}$$

(16.11)

which is the required condition of isochronism.

D'Alembert's principle states that the inertia forces and couples, and the external forces and torques on a body together give statical equilibrium.

Inertia is a property of matter by virtue of which a body resists any change in velocity.

$$\text{Inertia force } \mathbf{F}_i = -m \mathbf{f}_g \quad (13.1)$$

where

m = mass of body

$\mathbf{f}_g$ = acceleration of centre of mass of the body

The negative sign indicates that the force acts in the opposite direction to that of the acceleration. The force acts through the centre of mass of the body.

Similarly, an inertia couple resists any change in the angular velocity.

Inertia couple,

$$\mathbf{C}_i = -I_g \alpha \quad (13.2)$$

where I_g = moment of inertia about an axis passing through the centre of mass G and perpendicular to plane of rotation of the body

α = angular acceleration of the body

Let $\Sigma \mathbf{F} = \mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \text{ etc.}$ = external forces on the body

and $\Sigma \mathbf{T} = \mathbf{T}_{g1}, \mathbf{T}_{g2}, \mathbf{T}_{g3}, \text{ etc.}$ = external torques on the body about the centre of mass G .

According to D'Alembert's principle, the vector sum of forces and torques (or couples) has to be zero, i.e.,

$$\Sigma \mathbf{F} + \mathbf{F}_i = 0 \quad (13.3)$$

and

$$\Sigma \mathbf{T} + \mathbf{C}_i = 0 \quad (13.4)$$

These equations are similar to the equation of a body in static equilibrium, i.e., $\Sigma \mathbf{F} = 0$ and $\Sigma \mathbf{T} = 0$.

This suggests that first the magnitudes and the directions of inertia forces and couples can be determined, after which they can be treated just like static loads on the mechanism. Thus, a dynamic analysis problem is reduced to one requiring static analysis.

1. Hammer-blow

Hammer-blow is the maximum vertical unbalanced force caused by the mass provided to balance the reciprocating masses. Its value is $mr\omega^2$. Thus, it varies as a square of the speed. At high speeds, the force of the hammer-blow could exceed the static load on the wheels and the wheels can be lifted off the rail when the direction of the hammer-blow will be vertically upwards.

2. Variation of Tractive Force

A variation in the tractive force (effort) of an engine is caused by the unbalanced portion of the primary force which acts along the line of stroke of a locomotive engine.

If c is the fraction of the reciprocating mass that is balanced then

$$\text{unbalanced primary force for cylinder 1} = (1 - c) mr\omega^2 \cos \theta$$

$$\text{unbalanced primary force for cylinder 2} = (1 - c) mr\omega^2 \cos (90^\circ + \theta) = -(1 - c) mr\omega^2 \sin \theta$$

$$\text{Total unbalanced primary force or the variation in the tractive force} = -(1 - c) mr\omega^2 (\cos \theta - \sin \theta)$$

This is maximum when $(\cos \theta - \sin \theta)$ is maximum,

$$\text{or when } \frac{d}{d\theta} (\cos \theta - \sin \theta) = 0 \quad \text{or} \quad -\sin \theta - \cos \theta = 0 \quad \text{or} \quad \sin \theta = -\cos \theta \quad \text{or} \quad \tan \theta = -1$$

$$\text{or} \quad \theta = 135^\circ \text{ or } 315^\circ$$

$$\text{When} \quad \theta = 135^\circ$$

maximum variation in tractive force $= (1 - c) m r \omega^2 (\cos 135^\circ - \sin 135^\circ)$

$$= (1 - c) m r \omega^2 \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right) = -\sqrt{2} (1 - c) m r \omega^2$$

When $\theta = 315^\circ$

Maximum variation in tractive force $= (1 - c) m r \omega^2 (\cos 315^\circ - \sin 315^\circ)$

$$= (1 - c) m r \omega^2 \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) = \sqrt{2} (1 - c) m r \omega^2$$

Thus, maximum variation $= \pm \sqrt{2} (1 - c) m r \omega^2$

3. Swaying Couple

Unbalanced primary forces along the lines of stroke are separated by a distance l apart and thus, constitute a couple (Fig. 14.15). This tends to make the leading wheels sway from side to side.

Swaying couple = moments of forces about the engine centre line

$$= [(1 - c) m r \omega^2 \cos \theta] \frac{l}{2} - [(1 - c) m r \omega^2 \cos (90^\circ + \theta)] \frac{l}{2}$$

$$= (1 - c) m r \omega^2 (\cos \theta + \sin \theta) \frac{l}{2}$$

This is maximum when $(\cos \theta + \sin \theta)$ is maximum.

i.e., when $\frac{d}{dt}(\cos \theta + \sin \theta) = 0$ or $-\sin \theta + \cos \theta = 0$ or $\sin \theta = \cos \theta$ or $\tan \theta = 1$

or

$$\theta = 45^\circ \text{ or } 225^\circ$$

When $\theta = 45^\circ$, maximum swaying couple $= \frac{1}{\sqrt{2}} (1 - c) m r \omega^2 l$

When $\theta = 225^\circ$, maximum swaying couple $= -\frac{1}{\sqrt{2}} (1 - c) m r \omega^2 l$

Thus, maximum swaying couple

$$= \pm \frac{1}{\sqrt{2}} (1 - c) m r \omega^2 l$$

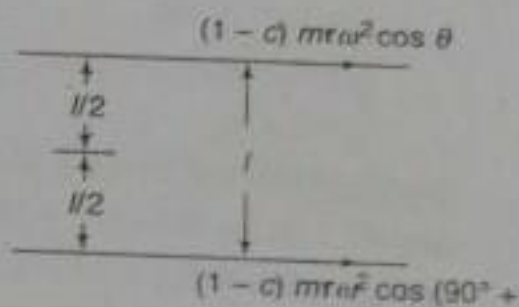


Fig. 14.15

(14.2)

(14.2)

2. V-Belt or Rope

In case of a V-belt or rope, there are two normal reactions as shown in Fig. 9.15 so that the radial reaction is equal to $2R \sin \alpha$.

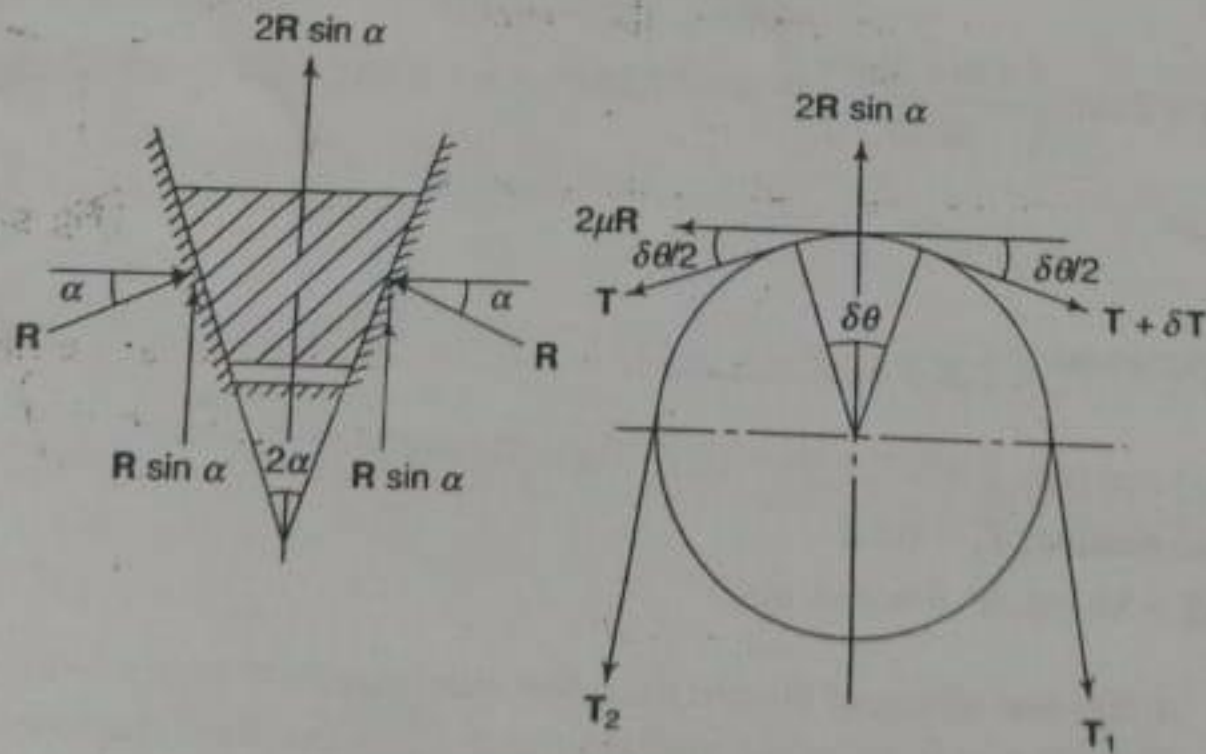


Fig. 9.15

Thus, total frictional force $= 2(\mu R) = 2\mu R$.

Resolving the forces tangentially,

$$2\mu R + T \cos \frac{\delta\theta}{2} - (T + \delta T) \cos \frac{\delta\theta}{2} = 0$$

For small angle of $\delta\theta$,

$$\cos \frac{\delta\theta}{2} = 1$$

$$\delta T = 2\mu R$$

(iii)

Resolving the forces radially,

$$2R \sin \alpha - T \sin \frac{\delta\theta}{2} - (T + \delta T) \sin \frac{\delta\theta}{2} = 0$$

As $\delta\theta$ is small,

$$\sin \frac{\delta\theta}{2} = \frac{\delta\theta}{2}$$

$$2R \sin \alpha - T \frac{\delta\theta}{2} - (T + \delta T) \frac{\delta\theta}{2} = 0$$

$$R = \frac{T \delta\theta}{2 \sin \alpha}$$

(iv)

or

From (iii) and (iv),

$$\delta T = 2\mu \frac{T \delta\theta}{2 \sin \alpha}$$

or

$$\frac{\delta T}{T} = \frac{\mu \delta\theta}{\sin \alpha}$$

Integrating between proper limits,

$$\int_{T_2}^{T_1} \frac{dT}{T} = \int_0^\theta \frac{\mu d\theta}{\sin \alpha}$$

or

$$\log_e \frac{T_1}{T_2} = \frac{\mu \theta}{\sin \alpha}$$

or

$$\frac{T_1}{T_2} = e^{\mu \theta / \sin \alpha}$$

(9.14)

The expression is similar to that for a flat-belt drive except that μ is replaced by $\mu/\sin\theta$, i.e., the coefficient of friction is increased by $1/\sin\theta$. Thus, the ratio T_1/T_2 is far greater in case of V-belts and ropes for the same angle of lap θ and coefficient of friction μ .

Again, it is to be noted that the above expression is derived on the assumption that the belt is on the point of slipping.

A chain is regarded in between the gear drive and the belt drive. Like gears, chains are made of metal and, therefore, occupy lesser space and give constant velocity ratios. Like belts, they are used for longer centre distances.

Advantages

- Constant velocity ratio due to no slip and thus, it is a positive drive.
- No effect of overloads on the velocity ratio.
- Oil or grease on surfaces does not affect the velocity ratio.
- Chains occupy less space as these are made of metals.
- Lesser loads are put on the shafts.
- High transmission efficiency due to no slip.
- Through one chain only, motion can be transmitted to several shafts.

Disadvantages

- It is heavier as compared to the belt.
- There is a gradual stretching and increase in length of chains. From time to time some of its links have to be removed.

- Lubrication of its parts is required.
- Chains are costlier as compared to belts.

It is seen that when a belt moves over the driving pulley, tension in the belt decreases from T_1 to T_2 .

Let l_1 = stretch in unit length of belt due to T_1

Let l_2 = stretch in unit length of belt due to T_2

Assuming that the strain in the belt is proportional to the stress in it,

As $T_1 > T_2$

Therefore, $l_1 > l_2$

Thus, a length $(1 + l_1)$ of belt approaches the driving pulley and a length $(1 + l_2)$ leaves it. As $(1 + l_1)$ is greater than $(1 + l_2)$, the belt slips back over the driving pulley. This slip is known as the *creep* of the belt.

On the driven pulley, the belt tension increases from T_2 to T_1 . This means a shorter length $(1 + l_2)$ approaches the driving pulley and a greater length $(1 + l_1)$ leaves it. Thus, the belt creeps forward by an amount $(l_1 - l_2)$. This makes the driven pulley to move at a slower speed than the belt.

Thus, the effect of creep is to slow down the speed of the belt on the driving pulley than that of the rim of the pulley and to reduce the rim velocity of the driven pulley than that of the belt on it. Therefore, the net effect of creep is to reduce the speed of the driven pulley than what it would have been without creep and thus, reducing the power transmitted.

Let σ_1 = stress on the tight side of the belt

σ_2 = stress on the slack side of the belt

ϵ_1 = strain on the tight side

ϵ_2 = strain on the slack side

N_1 = speed of the driving pulley

N_2 = speed of the driven pulley

l = original length of the belt

E = modulus of elasticity of the belt material

Assuming that the stress strain curve for the belt to be parabolic in nature,

$$\epsilon_1 = \frac{\sqrt{\sigma_1}}{E} \quad \text{and} \quad \epsilon_2 = \frac{\sqrt{\sigma_2}}{E}$$

Length on the tight side $= l + l\epsilon_1$

Length on the slack side $= l + l\epsilon_2$

$$\text{Velocity ratio} = \frac{V_2}{V_1} = \frac{\text{Peripheral speed of driven pulley}}{\text{Peripheral speed of driving pulley}}$$

The peripheral speed of a pulley is to be proportional to the approaching length of belt. Thus,

$$\frac{\pi D_2 N_2}{\pi D_1 N_1} = \frac{l + l\epsilon_2}{l + l\epsilon_1}$$

$$\frac{N_2}{N_1} = \frac{D_1}{D_2} \cdot \frac{1 + \epsilon_2}{1 + \epsilon_1} = \frac{D_1}{D_2} \left[\frac{E + \sqrt{\sigma_2}}{E + \sqrt{\sigma_1}} \right]$$

Some advantages of V-belts are

- Positive drive as slip between belt and pulley is negligible
- No joint troubles as V-belts are made endless
- Operation is smooth and quite
- High velocity ratio up to 10 can be obtained
- Due to wedging action in the grooves, limiting ratio of tensions is higher and the transmission
- Multiple V-belt drive increases the power transmission manifold
- May be operated in either direction with tight side at the top or bottom
- Can be easily installed and removed.

Disadvantages of V-belts are

- Cannot be used for large centre distances
- Construction of pulleys is not simple
- Not as durable as flat belts
- Costlier as compared to flat belts.

If it is desired that a belt transmits maximum possible power, two conditions must be fulfilled simultaneously.

1. Larger tension must reach the maximum permissible value for the belt.
2. The belt should be on the point of slipping, i.e., maximum frictional force is developed in the belt.

Now,

$$P = (T_1 - T_2)v = T_1 \left(T_1 - \frac{T_2}{T_1} \right) v = T_1 \left(1 - \frac{1}{e^{\mu\theta}} \right) v = T_1 kv$$

$$\text{where } k = 1 - \frac{1}{e^{\mu\theta}} = \text{constant}$$

$$\text{or } P = (T - T_c) kv = kTv - kmv^2 v = kTv - kmv^3$$

The maximum tension T in the belt should not exceed the permissible limit. Hence, treating T as constant and differentiating the power with respect to v and equating the same equal to zero, we get

$$\frac{dP}{dv} = kT - 3kmv^2 = 0$$

or

$$T = 3mv^2 = 3T_c$$

or

$$T_c = \frac{T}{3} \quad (9.18)$$

Therefore, for maximum power transmission, centrifugal tension in the belt must be equal to one-third of the maximum allowable belt tension and the belt should be on the point of slipping.

$$T_1 = T - T_c = T - \frac{T}{3} = \frac{2}{3}T$$

$$v_{\max} = \sqrt{\frac{T}{3m}}$$

The direction of the frictional force on the drum is to be opposite to that of its rotation while on the block it is in the same direction. Taking moments about the pivot O [Fig. 15.1(a)],

$$F \times a - R_n \times b + \mu R_n \times c = 0$$

$$R_n = \frac{Fa}{b - \mu c} \quad (15.2)$$

Also

$$F = R_n \frac{b - \mu c}{a} \quad (15.3)$$

- When $b = \mu c$, $F = 0$, which implies that the force needed to apply the brake is virtually zero, or that once contact is made between the block and the drum, the brake is applied itself. Such a brake is known as a *self-locking brake*.
- As the moment of the force F_f about O is in the same direction as that of the applied force F , F_f aids in applying the brake. Such a brake is known as a *self-energised brake*.

The difference between the maximum and minimum kinetic energies of the flywheel is known as the *maximum fluctuation of energy* whereas the ratio of this maximum fluctuation of energy to the work done per cycle is defined as the *coefficient of fluctuation of energy*.

The difference between the greatest speed and the least speed is known as the *maximum fluctuation of speed* and the ratio of the maximum fluctuation of speed to the mean speed is the *coefficient of fluctuation of speed*.

$$K = \text{coefficient of fluctuation of speed} = \frac{\omega_1 - \omega_2}{\omega}$$

The function of a governor is to maintain the speed of an engine within specified limits whenever there is a variation of load. In general, the speed of an engine varies in two ways—during each revolution (cyclic variation) and over a number of revolutions. In the former case, it is due to variation in the output torque of the engine during a cycle and can be regulated by mounting a suitable flywheel on the shaft. In the latter case, it is due to variation of load upon the engine and requires a governor to maintain the speed. The operation of a flywheel is continuous whereas that of a governor is more or less intermittent. A flywheel may not be used if there is no undesirable cyclic fluctuation of the energy output, but a governor is essential for all types of engines as it adjusts the supply according to the demand.

If the load on the shaft increases, the speed of the engine decreases unless the supply of fuel is increased by opening the throttle valve. On the other hand, if the load on the shaft decreases, the speed of the engine increases unless the fuel supply is decreased by closing the valve sufficiently to slow the engine to its original speed. The throttle valve is operated by the governor through a mechanism for the purpose.